# Yashwanth Piratla

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## Education

**University of California, Berkeley** **Expected May 2029**
*B.S. Mechanical Engineering, GPA: 3.95* *Berkeley, CA*
3D Modeling, Linear Algebra, Differential Equations, Physics 1, Data Structures

## Experience

**Liberty Robotics — FRC Team 4131** **Renton, WA**
*President* *Apr. 2022 – May 2026*

- Led a 40-member FIRST Robotics Competition team across robot development, mechanical design, manufacturing, outreach, and technical training.
- Coordinated mechanical, electrical, software, and media subteams through design reviews, prototyping, fabrication, testing, and competition preparation.
- Managed annual summer camp operations and outreach programs generating approximately $50K annually for team funding.

**Penn State Research** **Remote**
*Research Intern* *Jun. 2024 – Aug. 2026*

- Developed an iOS application using Swift and Apple HealthKit for wearable health data collection and analysis.
- Applied CNN, gradient boosting, and SVM models to wearable health telemetry and built the SwiftUI frontend.

**Verveba Telecom** **Dallas, TX**
*Engineering Intern* *Aug. 2024*

- Created test cases for a network organization tool used by U.S. telecommunications carriers.
- Analyzed telecom systems and assisted with patent development for a network optimization product in a Scrum/Jira workflow.

## Engineering Projects

**Three-Stage Cascading Elevator | *CAD, Mechanical Design, Manufacturing***

- Designed and assembled a three-stage cascading elevator with fully belt-driven rigging, combining vertical extension and arm pivot to reach a 7-ft-high algae scoring target.
- Optimized belt tensioning, pulley support, and stage synchronization through iterative prototyping, custom 3D-printed components, and aluminum fabrication.

**Deployable Multi-Level Climbing Mechanism | *CAD, Mechanical Design, FEA***

- Designed a deployable climbing mechanism using a dead-axle pivot, belt/gear transmission, chain drive, and carbon-fiber hooks to climb multiple bars.
- Balanced 115-lb weight, 12-in. extension, packaging, and structural constraints through FEA, prototyping, and extensive hook testing.

**MK4 Swerve Drivebase & Robot Architecture | *CAD, Packaging, Manufacturing***

- Designed frame rails, cross members, bumper mounts, battery mount, and bellypan architecture around an MK4 swerve drivetrain.
- Integrated an under-bumper intake and relocated electronics beneath the bellypan to improve subsystem packaging, wiring, and serviceability.

**42-Cell Battery Module Design | *Mechanical Design, FEA, Battery Systems***

- Designed a 42-cell 14S3P battery module, integrating BMS/Vsense hardware and achieving 50.4 V nominal and 14.85 Ah.
- Performed FEA under 20g vertical and 40g horizontal load cases and optimized cell spacing, structure, insulation, and manufacturing methods

## Honors & Awards

**DECA Washington State Champion** (2024, 2025) **FRC World Championship Qualifier** (2023, 2024) **FRC Sustainability Award** (2024)

## Technical Skills

**CAD & Mechanical:** Onshape, SolidWorks, Creo, Fusion 360, AutoCAD, Catia, Design, Robotics, Systems Integration, Pneumatics, Solenoids, DFM
**Manufacturing:** CNC Machining, Manual Machining, Fabrication, Precision Machining
**Software:** Python, MATLAB, Java, Swift, JavaScript, React, HTML, C++, MS Office Suite